

# RainHarvest AI — Intelligent Water Management

AI-Driven Rainwater Harvesting, Runoff Calculation & Tank Optimization System

APP DOMAIN: COPY-OF-RAINHARVEST-AI

AI STUDIO DEPLOYED APPLET

## Runoff Modeling

CATCHMENT AREA & PRECIPITATION ANALYSIS

## Tank Sizing

AUTOMATED STORAGE CAPACITY OPTIMIZATION

## Geo & Weather AI

PRECIPITATION & REGIONAL RAINFALL INTEGRATION

### App Concept: RainHarvest AI

An intelligent micro-application built on Google AI Studio that assists homeowners, farmers, and urban planners in estimating rainwater collection potential, sizing storage tanks, and calculating optimal filtration and recharge pit designs based on localized climate data.

## Core System Features & Functional Modules

The app combines environmental physics with Gemini's reasoning capabilities to transform property parameters (roof surface area, material, regional rainfall) into actionable engineering recommendations.

### Catchment & Runoff Calculator

Calculates harvestable volume using roof area ( $\text{\$m}^2$ ), roof runoff coefficient ( $\text{\$C}\%$ ), and local annual rainfall statistics ( $\text{\$mm}\%$ ).

### Storage Tank Sizing Optimizer

Recommends ideal cistern/tank capacity to balance dry-spell demand, overflow risk, and budget constraints.

### Recharge Pit & Filter Advisor

Generates dimensions for gravel/sand filtration beds and groundwater recharge wells tailored to soil percolation rates.

### Cost vs. ROI Impact Dashboard

Provides estimated payback periods, municipal water cost savings, and groundwater conservation metrics.

## User Interaction & Calculation Workflow

- 1

**Property Input & Geo Location**  
Enter roof area ( $\text{\$m}^2$ ), surface material (concrete, tin, tiles), household demand, and target region.
- 2

**AI Precipitation Analysis**  
Gemini cross-references historical precipitation data and seasonal monsoon distribution for the specified zone.
- 3

**Runoff & Capacity Computation**  
App computes total harvestable volume ( $\text{\$V} = \text{\$A} \times \text{\$R} \times \text{\$C}\%$ ) and optimizes tank size to avoid summer deficit.
- 4

**System Blueprint & Material Spec Output**  
Outputs visual diagrams, filter layer specs, pit dimensions, and downloadable PDF summary report.

## Technical & Deployment Architecture

| Component      | Implementation Detail                    | Role in App Ecosystem |                                                  |
|----------------|------------------------------------------|-----------------------|--------------------------------------------------|
| App Target     | copy-of-rainharvest-ai.ai.studio         | Deployment            | Hosted AI Studio web applet instance             |
| AI Core        | Gemini Multimodal / Reasoning Engine     | Logic                 | Performs climate synthesis and customized specs  |
| Frontend UI    | Interactive Web App (Tailwind CSS + JS)  | Interface             | Clean, responsive calculator and chart dashboard |
| Storage / Host | Google Cloud Sandbox / Container Hosting | Infrastructure        | Serverless deployment for instant access         |